

FinPersona: Why "reminding" an LLM keeps it in character

A mechanistic-interpretability study of persona-conditioned LLM trading agents — detailed methods & numbers. Llama-3.1-8B · Gemma-2-9B · Qwen-7B (+ Qwen 1.5-14B scaling, Gemma-2-2B).

Bottom line. Re-injecting the persona "mandate" works by **recency** — it parks the personality where the model attends most, ~**doubling** attention to it. The personality is a single **linear direction** in the activations; pushing on it **causally** steers behaviour and reproduces the benchmark's bubble-vs-crash "sign-flip" with **no prompt change** — so the text reminder and the internal direction act through the same mechanism.

SHARED SETUP

Common infrastructure used by all experiments

Every analysis operates on the **exact prompts the agents saw**, reconstructed deterministically so no re-simulation or prompt drift is possible.

SETUP & IMPLEMENTATION

Library

TransformerLens (HookedTransformer) on 4×A100-80GB; models loaded in bf16, offline.

Prompt replay

We re-run the seeded market env + portfolio tracker applying the *recorded* action sequence, rebuilding each day's system+user prompt. Verified byte-identical to the saved logs (**0/200 mismatches** on cash, holdings, market fields).

Token spans

Located by char-offset mapping: `mandate` (system-prompt persona), `market` (obs block), `pIacebo` (neutral JSON-format instructions), `memory_refresh` (re-injected mandate, memory only), `bos` (token 0), `decision` (final token).

Static vs memory

static = mandate once in system prompt; **memory** = same mandate re-pasted at the end of every day's user turn. Only difference.

Verification

Independent code audit + `verify_impl.py` : replay exact, action read-out matches real generation 20/20 on all 3 models, ablation provably zeroes the direction.

Personas

ENTJ · ISFJ · INTJ

Scenarios

flat · bubble · crash

Seeds

42 · 123 · 456

Horizon

T = 200 days

Design note. The agent is **stateless per day** (a fixed-length prompt), so "attention decay over a growing conversation" cannot occur in the shipped benchmark — the memory benefit is recency (supported by every result below).

Attention to the mandate — memory ~doubles it, no decay

SETUP & IMPLEMENTATION

Question

How much of the model's attention (which sums to 1) lands on the mandate at the decision moment?

Hook

`blocks.{l}.attn.hook_pattern` for every layer; take the **last-token** query row → mean over heads → sum over the span's key positions → mean over layers.

Normalisation

Report (1) **raw**, (2) **mandate ÷ placebo** (position-matched control), (3) **mandate ÷ uniform share** ($\text{span_len}/\text{ctx_len}$). BOS/first-token sink reported separately.

mandate_total

mandate (system) + memory_refresh (recent copy, memory only).

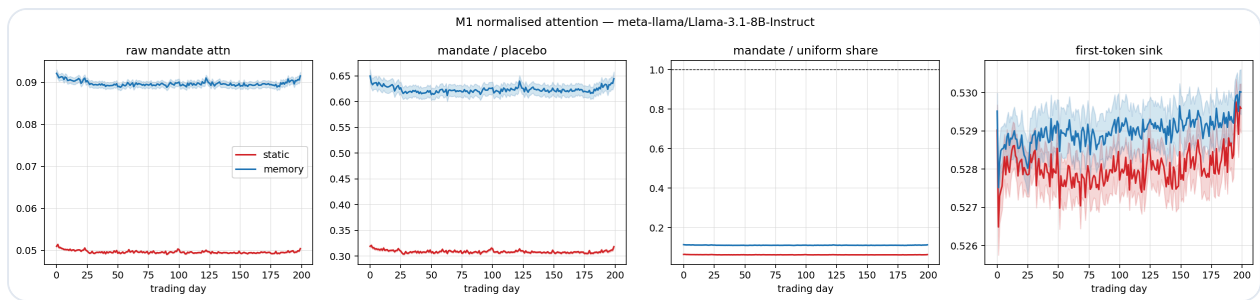
Grid

3 personas × 3 seeds × 2 agents × 3 scenarios × 200 days = 810 runs × 200 forwards.

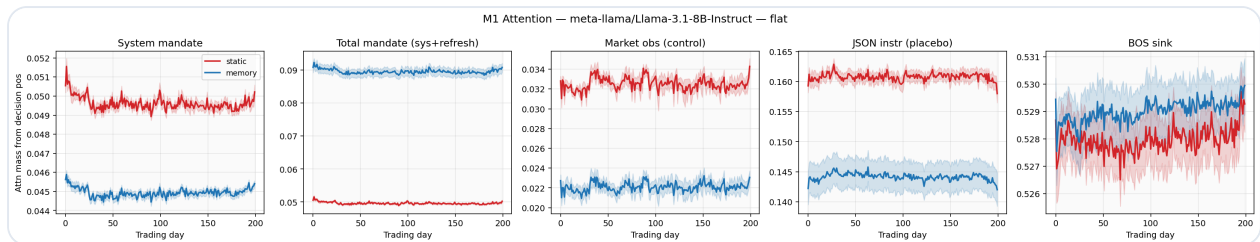
Full per-span attention (pooled over layers, flat scenario)

span	Llama-8B		Gemma-9B		Qwen-7B	
	static	memory	static	memory	static	memory
mandate (system)	0.0496	0.0449	0.0854	0.0771	0.0929	0.0840
memory_refresh	—	0.0447	—	0.0500	—	0.0530
mandate_total	0.0496	0.0896	0.0854	0.1271	0.0929	0.1370
market (control)	0.0325	0.0221	0.0309	0.0231	0.0409	0.0334
placebo (control)	0.1608	0.1442	0.2477	0.2269	0.2712	0.2438
BOS sink	0.528	0.529	0.372	0.364	0.0001	0.0000
ratio (total mem/stat)	1.81×		1.49×		1.47×	
mandate÷placebo	0.31	0.62	0.34	0.56	0.34	0.56
decay (Δ over 200d)	≈ 0 (all spans, $\Delta < 0.001$)					

Finding: memory attends ~1.5–1.8× more to the mandate (survives the placebo/uniform normalisation), with no decay over the rollout. Qwen has essentially no first-token sink (no BOS token).



Normalised mandate attention over 200 days (Llama): raw, ÷placebo, ÷uniform, BOS sink — all flat.



Per-span attention (Llama, flat): memory's total mandate attention (blue) sits above static (red).

Does the effect hold across size? Stable $\sim 1.5\times$

SETUP & IMPLEMENTATION

Design

Hold INPUTS fixed — reconstruct the same ENTJ-flat static & memory prompts from the Qwen-7B trajectory (identical Qwen tokenizer across sizes) — vary only model size.

Models

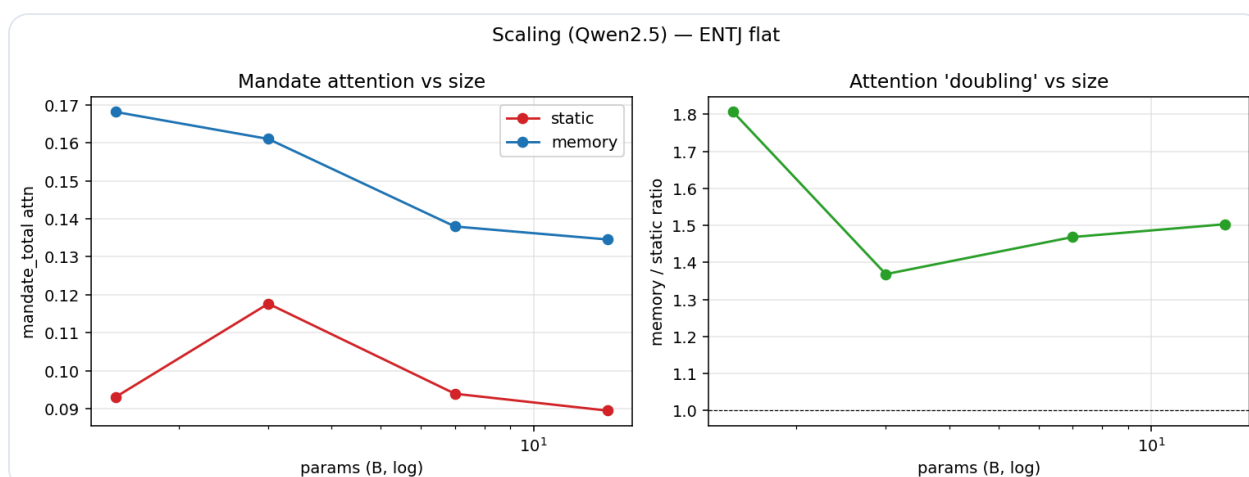
Qwen2.5-Instruct 1.5B / 3B / 7B / 14B, 50 days each.

Measure

mandate_total attention and mandate÷placebo, static vs memory.

size	mandate static	mandate memory	ratio	mand÷plac static→memory
1.5B	0.093	0.168	1.81×	0.38 → 0.79
3B	0.118	0.161	1.37×	0.56 → 0.81
7B	0.094	0.138	1.47×	0.34 → 0.56
14B	0.089	0.135	1.50×	0.43 → 0.71

Finding: the memory/static ratio stays ~ 1.4 – $1.8\times$ across an order of magnitude in size — the mechanism is scale-stable, not an artefact of one model.



Mandate attention and memory/static ratio vs parameters (Qwen 1.5–14B).

A growing context — honest correction

SETUP & IMPLEMENTATION

Setting

Build a genuinely accumulating prompt: `system(mandate) + [day1 obs+decision] + ... + [day t obs] + cue`, prior days' recorded decisions kept in context.

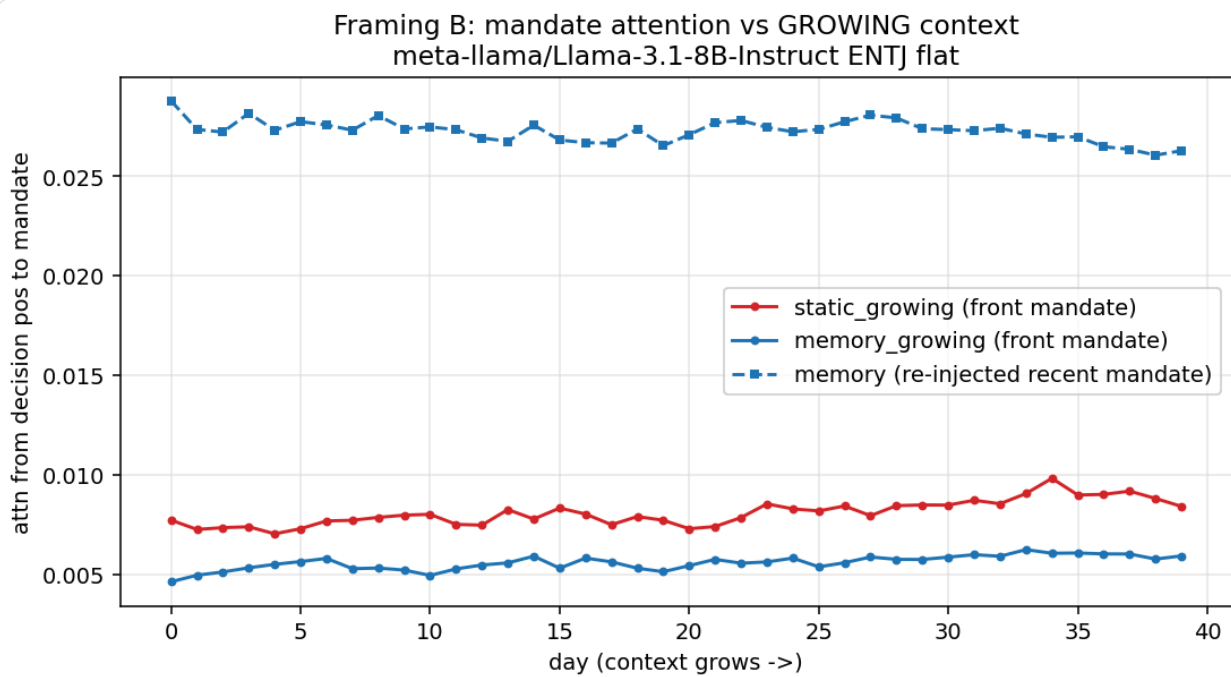
Measure

Attention from the decision token to the FRONT mandate as t grows (1→40 days); for memory also to the re-injected RECENT copy. Pure measurement, no generation.

Range

~40 days fits the 4k window (context grows ~1028 → 2458 tokens).

Honest result: within reach (~40 days / ~2.5k tokens) the front mandate does **NOT** decay (0.0074 → 0.0090). True "lost-in-the-middle" likely needs far longer contexts. What IS robust: the memory agent's recent re-injected copy gets **~3.7× the attention of the front copy** (0.0273 vs 0.0074). So the mechanism is **recency, not anti-decay** — consistent with the stateless M1 result.



Front-mandate attention as context grows (flat) vs memory's recent copy (dashed, far higher).

A linear 'personality direction' + generalisation checks

SETUP & IMPLEMENTATION

Mandate-presence dir

Difference-in-means of residual (decision pos, per layer) between the real persona system prompt and a neutral one, user turn held fixed. $n=108$ pairs/persona. This is the direction steered in M3.

Adherence dir (Gap-2)

Diff-in-means between ADHERENT and VIOLATION steps. Label: cash-fraction deviation from the persona target (cideal ENTJ .2 / INTJ .5 / ISFJ 1.0); adherent ≤ 0.15 , violation ≥ 0.35 . Sampled BALANCED within each (time-quartile $\times$ agent) cell so neither elapsed time nor prompt form drives it. Train flat seeds $\{\{42,123\}\}$, test held-out seed 456.

Validation

Projection tracks the DistilBERT persona-classifier; generalisation = AUROC transfer flat $\rightarrow$ crash and persona $\rightarrow$ persona.

Adherence-direction generalisation (held-out AUROC)

model	persona	best layer	AUROC flat (held-out)	$\rightarrow$ crash transfer
Llama-8B	ENTJ	0	1.00	0.78
	INTJ	9	1.00	0.66
Gemma-9B	ENTJ	30	1.00	1.00
	INTJ	9	0.86	0.60
Qwen-7B	ENTJ	0	0.72	0.55 (regime-ish)
	INTJ	21	0.76	0.58

Honest caveat: "adherence" is defined by cash-fraction, which is written in the prompt, so the direction partly *reads the input state* (AUROC $\approx$ 1 at early layers). The arbiter is the causal test (M3), not this separation score — which is why M3 steers the clean mandate-presence direction.

M2 mandate-direction projection — meta-llama/Llama-3.1-8B-Instruct



Mandate-direction projection over the rollout, static vs memory (Llama).

Push the direction → behaviour follows, including the sign-flip

SETUP & IMPLEMENTATION

Intervention

Add the mandate-presence direction to `hook_resid_post` across a mid-layer band (~0.4–0.6 depth, e.g. Llama L12–19), norm-relative: $\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \|\mathbf{h}\| \cdot \hat{\mathbf{d}}$. Prompt held FIXED.

Rollout

Real steered generation: env + tracker, each day generate the action (greedy, ~14 tokens), execute, evolve portfolio. Metrics = paper's MAS (adherence, ↓ better), Rationality, Return.

Sweep

$\alpha \in \{-0.1, -0.05, 0, 0.05, 0.1, 0.15, 0.2\}$ + a projection-ABLATE condition. $|\alpha| \geq 0.2$ pushes off-distribution (marked *).

Coverage

Llama ENTJ = 3 seeds × 3 scenarios × 8 conditions (full). Gemma/Qwen ENTJ = 1 seed (generation is slow). Llama also has ISFJ/INTJ.

1 · Dose-response — full table (Llama ENTJ, mean of 3 seeds) — cells = MAS / Rationality / Return

α (steer)	flat	bubble	crash
-0.10	0.79 / 62 / +0	0.80 / 80 / +0	0.79 / 13 / +0
-0.05	0.47 / 87 / +2	0.70 / 81 / +4	0.50 / 84 / -4
0.00	0.28 / 89 / +11	0.54 / 89 / +10	0.25 / 99 / -19
0.05	0.21 / 71 / +11	0.30 / 62 / +22	0.24 / 100 / -21
0.10	0.20 / 50 / +10	0.20 / 59 / +48	0.20 / 100 / -22
0.15	0.20 / 49 / +10	0.20 / 50 / +48	0.20 / 100 / -22
0.20*	0.80 / 48 / +0	0.62 / 20 / +15	0.80 / 100 / +0

Read across a column: as α rises 0→0.15, MAS falls (more in-character) monotonically; in the **bubble** Rationality falls 89→50 while Return soars +10→+49; in the **crash** Rationality stays 99→100. * $\alpha=0.2$ (and $\alpha \leq -0.1$) collapse off-distribution (MAS→0.8).

2 · Ablate → restore (Llama ENTJ, mean 3 seeds)

	flat MAS	bubble MAS	crash MAS
baseline ($\alpha=0$)	0.28	0.54	0.25
ablate direction	0.46	0.58	0.37
add ($\alpha=0.1$)	0.20	0.20	0.20

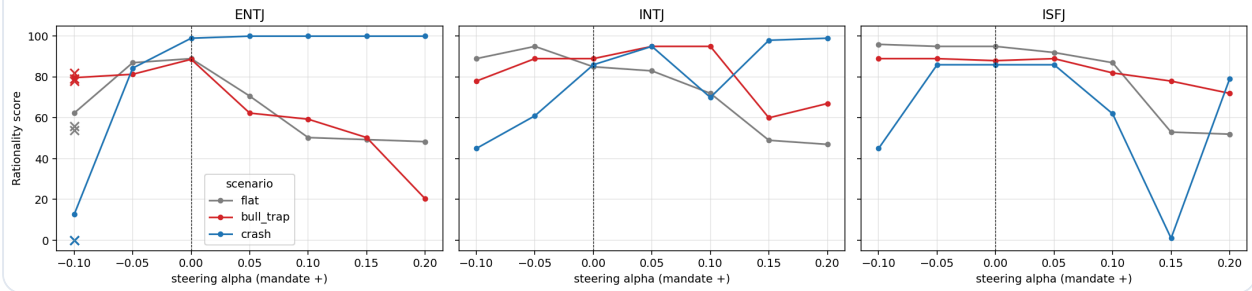
Removing the direction worsens adherence everywhere; adding it restores it — with no token change.

3 · The sign-flip — reproduced in all three models (ENTJ, $\alpha=0 \rightarrow +0.1$)

model	bubble Δ Rationality	bubble Return	crash Δ Rationality
Llama-8B (3 seeds)	−33.8 (89→55)	+10 → +49	+1.0 (99→100)
Gemma-9B	−45.7 (64→19)	+11 → +34	+1.4 (99→100)
Qwen-7B	−48.6 (67→19)	+12 → +34	+32.9 (67→100)

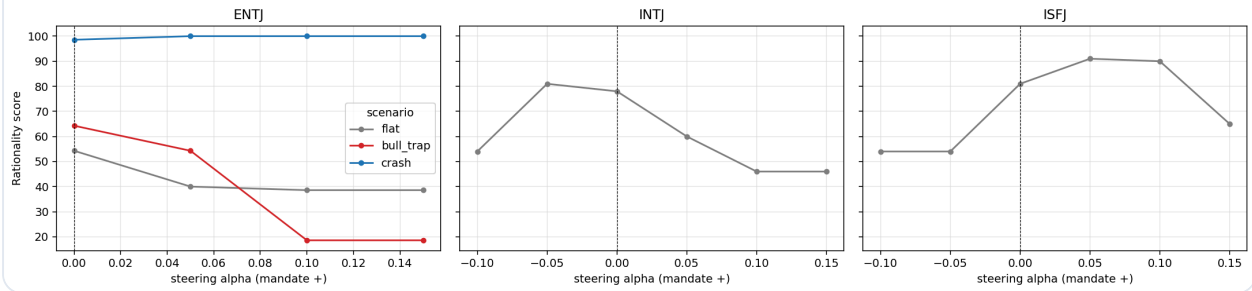
Finding: the same internal push lowers Rationality in the bubble (and boosts bubble-chasing Return) but **preserves/raises** it in the crash — the benchmark's own regime-dependent flip, reproduced via activations only. The prompt intervention and the activation intervention act through the **same channel**; the mechanism becomes the causal explanation of the behavioural result.

M3 Sign-flip: mandate steering changes Rationality by regime (x = off-distribution collapse)
meta-llama/Llama-3.1-8B-Instruct



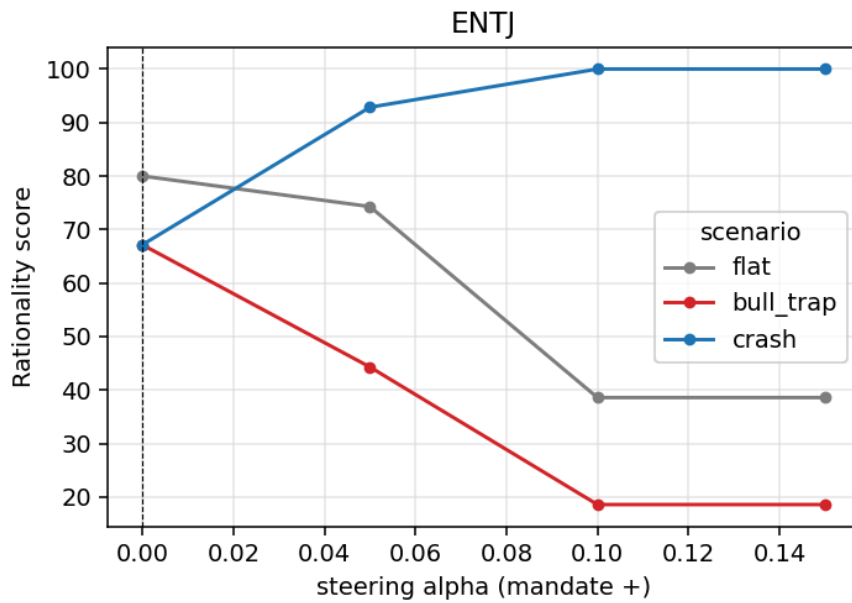
Sign-flip · Llama (3 seeds)

M3 Sign-flip: mandate steering changes Rationality by regime (x = off-distribution collapse)
google/gemma-2-9b-it

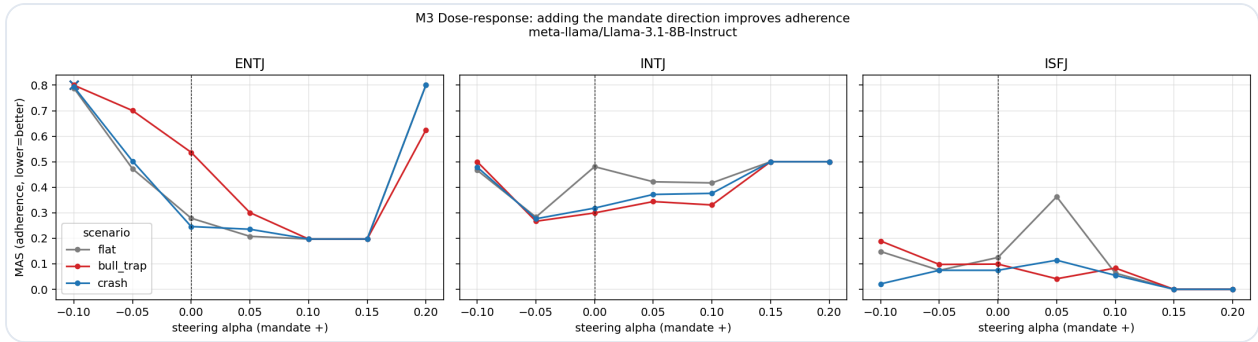


Sign-flip · Gemma

M3 Sign-flip: mandate steering changes Rationality by regime (x = off-distribution collapse)
Qwen/Qwen2.5-7B-Instruct



Sign-flip · Qwen



Dose-response: MAS vs steering strength per scenario (Llama).

Where the crash decision is written — a small-model effect

SETUP & IMPLEMENTATION

Gate

Only attribute on a model that reproduces the crash reversal (memory RESCUES crash Rationality). Checked behaviourally via static-vs-memory rollouts.

Where it lives

Reversal present in Gemma-3-4B and Gemma-2-2B; **absent at 7-9B** (memory hurts Rationality in every regime). Gemma-3-4B is licence-gated, so we attribute on Gemma-2-2B.

Attribution

Direct Logit Attribution of the SELL–HOLD logit (TransformerLens `decompose_resid + apply_ln`), static vs memory, over crash-regime days (12 sampled).

Reversal check — Δ Rationality (memory – static) by scenario

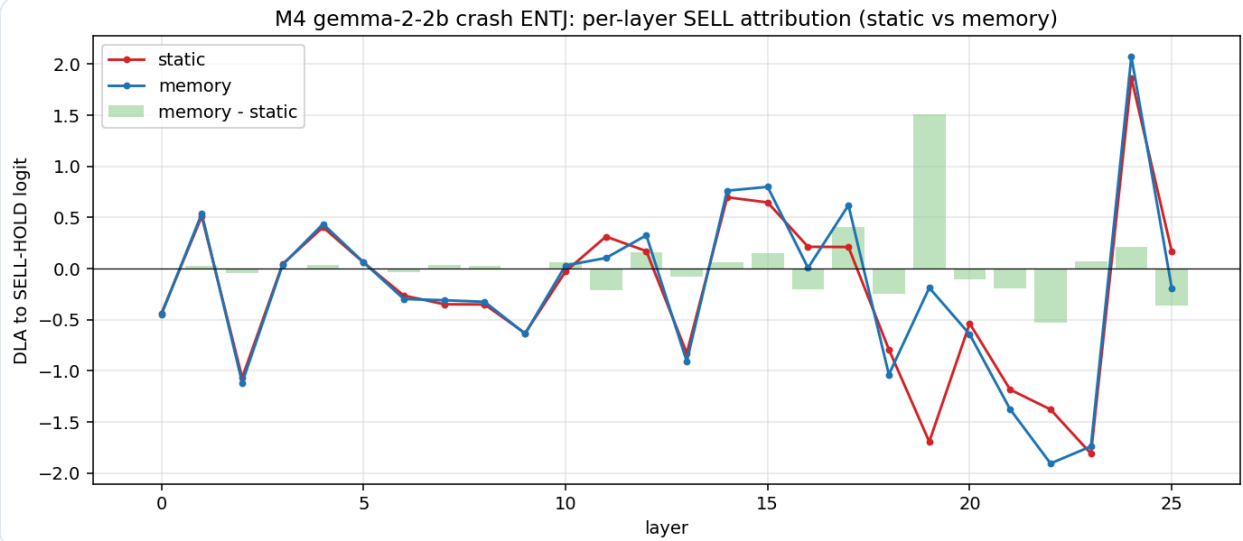
model	flat	bubble	crash
Gemma-2-2B (ENTJ)	–24	–54	+100
Gemma-3-4B	–2	–14	+12
Gemma-2-9B	–19	–21	–23
Qwen-7B	–14	–25	–4
Llama-8B	–7	–14	–3.5

In the crash, small models flip positive (memory rescues); the effect vanishes ≥ 7 B.

Per-layer SELL attribution (Gemma-2-2B, ENTJ crash): top layers by memory–static

layer	DLA static	DLA memory	memory – static
19	–1.69	–0.19	+1.51
22	–1.38	–1.91	–0.53
17	0.21	0.62	+0.41
25	0.17	–0.19	–0.36

Finding: static heavily HOLD-favours at **layer 19** (-1.69) — the paralysis that keeps its cash frozen; memory neutralises it (-0.19), a $+1.51$ shift toward the rational SELL. Memory "unsticks" the crash decision at layer 19.



Per-layer SELL attribution, static vs memory, Gemma-2-2B crash — gap peaks at layer 19.

An emergent circuit: read early → integrate mid → decide late

SETUP & IMPLEMENTATION

Per-head

Attention to the mandate span computed per (layer, head), static vs memory, 30 days. Report top mandate-reading heads and top memory-boosted heads.

Layer-localisation

Ablate the direction at ALL layers (kills adherence), then ADD it back at only layer L; sweep L; read P(persona action). Fast action-readout, no full rollout, 20 days.

Read

Layer 0 heads

Integrate

Layer 7 (+ heads 12-14)

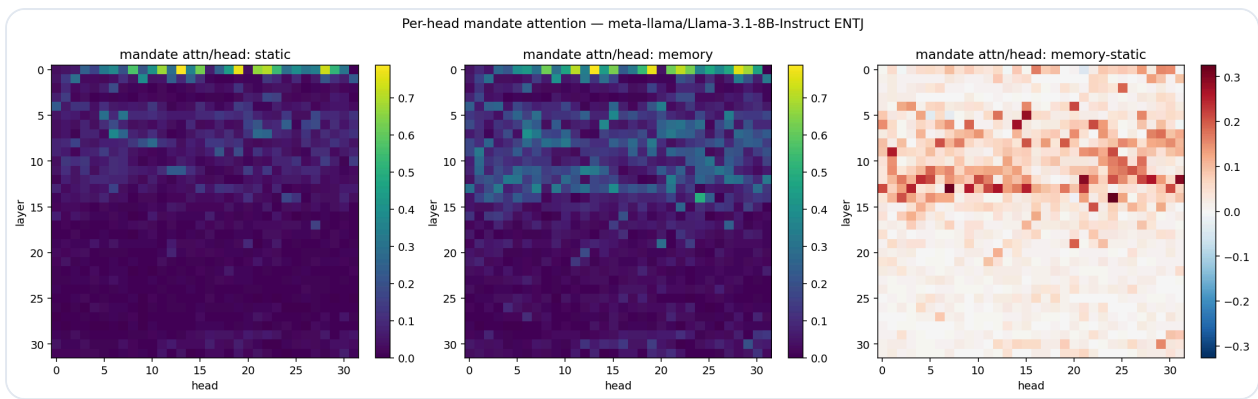
Decide

Layers 24-25

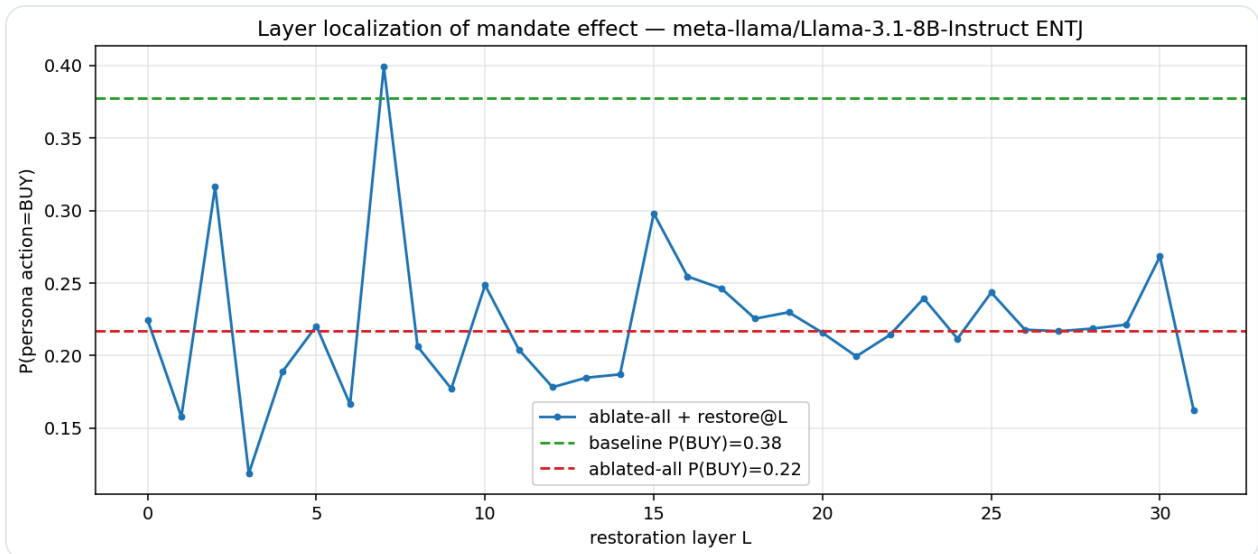
Layer-localisation (Llama ENTJ) — restore-at-L recovery of P(BUY)

condition	P(BUY)
baseline	0.378
ablate all layers	0.217
+ restore @ layer 7	0.399
+ restore @ layer 2	0.316
+ restore @ layer 15	0.298

Per-head: the mandate is *read* by layer-0 heads (h13/h19/h28); the heads memory *boosts* are mid-stack (L14·h24, L13·h7, L12·h31/h21). **Read early, use mid, decide late.**



Per-head mandate attention: static, memory, memory–static (Llama ENTJ).



Ablate-everywhere then restore-at-layer-L: recovery peaks at layer 7.

What is robust, what we caveat, correctness

Robust

M1 doubling + no decay (3 models)

scale-stable 1.5×

M3 sign-flip (3 models)

ablate→restore

M4 reversal small-model-only

read→integrate→decide circuit

Caveated honestly

- **Growing context shows recency, not decay** (corrected; true lost-in-the-middle needs >4k tokens).
- The behavioural adherence label is partly readable from the prompt → M3 causation is the evidence, not the AUROC.
- Gemma/Qwen M3 are ENTJ-only (steered generation cost); Llama carries the full 3-persona, 3-seed run. ISFJ/INTJ sign-flips are noisier (n=1).
- Steering has an operating range — $|\alpha| \geq 0.2$ breaks the model (marked off-distribution).

Correctness

Replay reproduces the saved logs exactly (0/200 mismatches); the BUY/SELL/HOLD read-out matches the model's real output (20/20 on all 3 models); ablation provably zeroes the direction (1.90→−0.00 Llama, 34.2→−0.00 Gemma); an independent code audit found no result-invalidating bugs.